SPARSE RECOMPUTATION AND BOUNDARY PLACEMENT IN TRANSCRIPT SEGMENTATION: EXPLORATORY NEGATIVE RESULTS ON YTSEG AND A PREREGISTERED TEST ON VT-SSUM

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ABSTRACT

Rerunning a chaptering model on a growing transcript is costly. We asked whether a cheap trigger can decide when to recompute a causal transcript segmentation model under a fixed call budget, and whether the value of the model's boundary placement holds beyond its training benchmark. On English YTSeg transcripts, using a reimplemented online MiniSeg with a 12.5% call budget, a strictly causal learned trigger reached a macro-averaged collar-1 recall of 0.171, against 0.472 for full recomputation. Its recall ratio (bootstrap intervals spanning 0.347 to 0.377 across seeds) fell below the target of 0.90, and its gain over matched random calls (0.051 to 0.075) fell below the target of 0.08. Post-hoc foresight diagnostics show the ratio target was barely reachable: 0.80 with knowledge of the labels, 0.91 with knowledge of both the labels and the model's hits, and 0.54 for a perfect proposer under the pacing controller we used. These YTSeg results are exploratory because the test split had already been opened. With segment counts held equal, MiniSeg placement beat even sentence spacing by 0.295 in collar-1 F1. We preregistered a test of this result on VT-SSum, a lecture dataset with slide-derived labels, with no retraining or tuning (958 test videos). The advantage over even spacing fell to 0.056 (0.044 to 0.067), inconclusive against the threshold of 0.05. It reversed at collars 2 and 3. The advantage over random placement was supported, and the claim that placement matters more than count was falsified, because the uncalibrated model predicted only 2,429 of 32,600 boundaries when it chose its own count. Exploratory peak suppression raised the collar-1 advantage to 0.119 and removed the reversal at its wider setting, so that reversal reflects how top-K boundaries were selected as well as the tolerance. Reported rankings therefore depend on selection rule, tolerance, and calibration, and results should not be assumed to transfer to differently defined segments.

Keywords transcript segmentation, video chaptering, sparse recomputation, boundary evaluation, negative results

1. Introduction

Automatic chaptering divides a long video into navigable semantic units. Some methods treat it as text segmentation over a transcript (Retkowski and Waibel, 2024), and others work in the text domain with transcripts and frame captions (Ventura et al., 2025). For a live or growing stream, rerunning such a model on every new sentence is costly, which motivates two practical questions.

The first is about compute. If a cheap signal could tell a system when its expensive model needs to run, much recomputation might be skipped with little loss. Event-triggered invocation is an active idea in streaming video and language systems (Guo et al., 2026; Wang, 2026). We tested it for transcript boundary detection under an explicit, per-video call budget.

The second is about evaluation. Boundary F1 depends on how many segments are predicted and on the tolerance used for matching. A simple baseline can look strong for these reasons alone. We therefore separated placement from count by giving every method the same number of boundaries, and then tested whether the result held on a dataset the model had never seen.

Our contributions are:

1. A causal, exact-budget evaluation of sparse recomputation on YTSeg, with a negative result and foresight diagnostics that separate the limits of the budget, the controller, and the signal (exploratory).
2. A count-held-equal comparison of placement across tolerances on YTSeg (exploratory).
3. A preregistered external test on VT-SSum, in which the advantage over even spacing shrinks and reverses at wider tolerances, plus an exploratory analysis showing how much of that reversal depends on the selection rule.
4. A record of how instrument errors arose in an autonomous research system and changed its conclusions (Section 6.3).

We make no claim about Korean content, videos longer than 30 minutes, audio or visual cues, or deployability.

2. Related Work

Transcript segmentation and chaptering. YTSeg and MiniSeg (Retkowski and Waibel, 2024) introduced a benchmark of spoken YouTube content and an efficient hierarchical segmentation model, and are our anchor. That paper also evaluates online segmentation, including a condition with no future context, and cross-dataset transfer (YTSeg to QMSum zero-shot F1 of 9.99), so limited transfer is not a new observation. Later work applies MiniSeg to other spoken benchmarks (Retkowski and Waibel, 2025). Chapter-Llama (Ventura et al., 2025) addresses hour-long videos by giving speech transcripts and selected frame captions to a large language model. VidChapters-7M (Yang et al., 2023) provides user-annotated chapters at large scale. VT-SSum (Lv et al., 2021) and AVLectures (Singh S et al., 2023) focus on lectures.

Audio chaptering. Retkowski et al. (2026) compare text-based models with acoustic features and audio-only models. They train MiniSeg on YTSeg, report limited transfer of an audio model to meetings, and find rankings consistent across 6 to 12 second chunk sizes. Their protocol differs from ours, so that consistency is not a direct contradiction of our tolerance results. We did not evaluate audio.

Online boundary detection and event-triggered processing. Online Generic Event Boundary Detection (Jung et al., 2025) detects event boundaries in streaming video without future frames. Event-VStream (Guo et al., 2026) triggers language generation at detected event boundaries. Wang (2026) treats the decision of when to invoke a language model as a sequential stopping problem, gives formal results, and compares triggers with periodic and random calling on sensor streams, concluding that invocation timing matters more than frequency. Our transcript results show a weaker gain from timing. Beyond the different domain and task, our setting differs in using an exact per-video budget, a transcript segmenter, and label-aware foresight diagnostics.

Budgeted attention to boundaries. B-ACT (Helvaci and Cheung, 2026) allocates annotation budget to boundary regions in temporal action segmentation. It addresses labeling rather than inference, but shares the premise that a limited budget should go where boundaries are.

Novelty. Our protocol combination is narrow rather than a new research direction. A limited search found no prior paper combining budgeted triggers with transcript segmentation, but provider rate limits prevented complete coverage, so this is not evidence of absence.

3. Data and Model

3.1 Datasets

	YTSeg	VT-SSum
Content	Spoken English YouTube transcripts	Lecture transcripts (VideoLectures.NET)
Revision	`4ee1c1fccc5a9e118ff0918f3df1e61aadca8970`	`b72da9e4009e2bf1a54d858fd0d1dbf434ac1d12`
License	CC BY-NC-SA 4.0	CC BY-NC-ND 4.0
Train / validation / test	16,404 / 1,447 / 1,448	7,692 / 962 / 962
Labels	Chapter boundaries	Sentence groups derived from slides
Test sentences	289,936	294,123 (after exclusions)
Test boundaries	11,848	32,600 (after exclusions)
Mean / median segment length (sentences)	21.8 / 11	8.8 / 5
Segments of 3 sentences or fewer	9.0%	35.6%
Boundary rate per sentence	4.1%	11.1%

We used text only and did not download audio or video. VT-SSum was used locally and is not redistributed.

3.2 Model

We reimplemented MiniSeg from the paper because we found no public official implementation: a fine-tuned all-MiniLM-L6-v2 sentence encoder (Wang et al., 2020; Reimers and Gurevych, 2019) with mean pooling, a 12-layer RoPE transformer document encoder (384 dimensions, 8 heads, dropout 0.1), and a per-sentence binary head. Training used weighted cross-entropy (weights 1 and 2), AdamW at a learning rate of 2.5e-5 with warmup and a cosine schedule, batches of about 115,000 tokens, and 15 epochs. As in the paper, encoder gradients were sampled for half of the documents. We selected the best epoch on validation F1. Online mode uses a causal attention mask.

We trained one seed only. Unlike the paper, sentence-encoder backpropagation was also skipped for documents over 40,000 tokens because of GPU memory.

Table 1: Model anchor on the YTSeg test split (exact sentence-level metrics).

Model	F1	Precision	Recall	Pk
Ours, offline	47.05 ± 0.78	52.79	42.43	28.36
Ours, online	44.26 ± 0.77	50.84	39.20	29.88
Retkowski and Waibel (2024), YTSeg-only offline	43.37 ± 0.60	45.44 ± 0.83	41.48 ± 0.85	28.73 ± 0.39

Pk is the segmentation error metric of Beeferman et al. (1999). Our offline result is 3.68 points above the published row. Because the implementations and training conditions are not shown to be identical, we treat this as a comparable baseline, not a strict reproduction. The paper gives no YTSeg-only online row; its online rows use Wiki-727K pretraining. Unless stated otherwise, the results below use the online model.

4. Methods

4.1 Sparse recomputation (YTSeg, exploratory)

A controller may reveal the model's output on a limited budget: r = 0.125 calls per sentence, with an identical count K_v per video in every sparse arm. Each call at sentence t reveals the two-position window {t-1, t} (W = 2), and calls must be at least LAT = 6 sentences apart. The warm-start call at sentence 0 counts toward the budget. Policies see only the current cheap sentence embedding, the sentence length, and model state cached from their own most recent actual call.

Calls are also paced: a controller may not run ahead of a near-uniform schedule (`pace_cap`) and is forced to call when it falls behind (`pace_floor`). This guarantees the exact budget, but it restricts how far a good signal can concentrate calls, as Section 5.2 shows.

Arms: a learned probe, an embedding-only probe, even calling, random calling, a causal length rule, and full recomputation as the reference. Probes and the length threshold were fitted on validation only. A hash manifest of code and calibration was written before test data were loaded. Metrics in this subsection are macro-averaged per video; videos with no predictions score zero precision.

Criteria, at sentence collar 1: **C1** recall of at least 0.90 of full recomputation, and **C2** a recall gain of at least 0.08 over random calling. Each counts as falsified when every seed's interval upper bound falls below the threshold.

Runner-61 gave a label-aware oracle the same budget and measured reference coverage. A follow-up tested whether its coverage falls sharply in the densest quartile (validation-derived cut points), adjusting for minimum boundary spacing, collar overlap, duration, speaker composition, boundary count, eligible sentence count, and K rounding. These variables are largely determined by density, so this adjustment may be conservative.

After review, we added post-hoc foresight diagnostics that pass schedules through the online model's output and score predicted recall: an oracle that knows labels, an oracle that also knows where the model is correct, and a perfect label-aware proposer inside the pacing controller. None can be deployed.

4.2 Placement at equal count

Within a video, all arms emit the same K boundaries. Placement arms: **MiniSeg** (the K largest logit margins, with no peak suppression), **cohesion** (the K largest adjacent-embedding dissimilarities), **even** (evenly spaced by sentence index), and **random**. Count policies: oracle K (the reference count) and estimated K (the count of MiniSeg positive predictions).

Artifacts from the cycle name the even arm "equal duration". It is even spacing by sentence index, not by time, and we use the accurate name here.

We report micro F1 with greedy one-to-one nearest matching at sentence collars 0 to 3, and paired video bootstrap intervals (2.5% to 97.5%, 1,000 resamples) for five seeds. Seeds change only the bootstrap and the random arm; they are not model retraining replicates. For deterministic arms, the range of intervals across seeds reflects only bootstrap Monte Carlo noise, and it is not itself a confidence interval.

4.3 Preregistered test on VT-SSum

Before reading any VT-SSum test label, we froze a preregistration and the analysis script by SHA-256. The model was not retrained or tuned. A claim is **supported** if the lowest interval lower bound across five seeds exceeds its threshold, **falsified** if the highest upper bound falls below it, and otherwise **inconclusive**.

ID	Quantity (oracle K, online model)	Threshold
P1	Collar 1: MiniSeg minus even	0.05
P2	Collar 1: MiniSeg minus random	0.05
P3	MiniSeg minus even at collar 0, minus the same at collar 3	0
P4	Collar 1: (MiniSeg minus even) minus (MiniSeg at oracle K minus MiniSeg at estimated K)	0

As an instrument check, the same script on YTSeg online outputs reproduced earlier collar-1 values (MiniSeg 0.551, cohesion 0.339, even 0.256). Four test videos were excluded by the prespecified rule; all had a single segment.

Deviations and disclosures: (1) the author read the VT-SSum README and part of one development file before writing the preregistration; (2) development results for the online model were seen before the freeze, with P1 supported there (lower bound 0.051) and P4 already falsified, but no threshold or arm changed; (3) the primary model was changed from offline to online before the freeze, when the instrument check showed that the earlier results had used online outputs; (4) the offline development split ran after the online test, with no effect on the frozen analysis; (5) the preregistration lists the offline model as having no verdict, but below we also state what the rule would give for it; (6) quartile effects use one bootstrap seed, which the preregistration did not specify; (7) hypotheses and thresholds came from the already opened YTSeg results; (8) the analyst and the drafter were the same AI agent. No correction was made for multiple comparisons; P2 to P4 sit far from their thresholds.

Analyses after the freeze that were not preregistered (`confirmatory_vtssum/posthoc_review_checks.py`) are labeled exploratory.

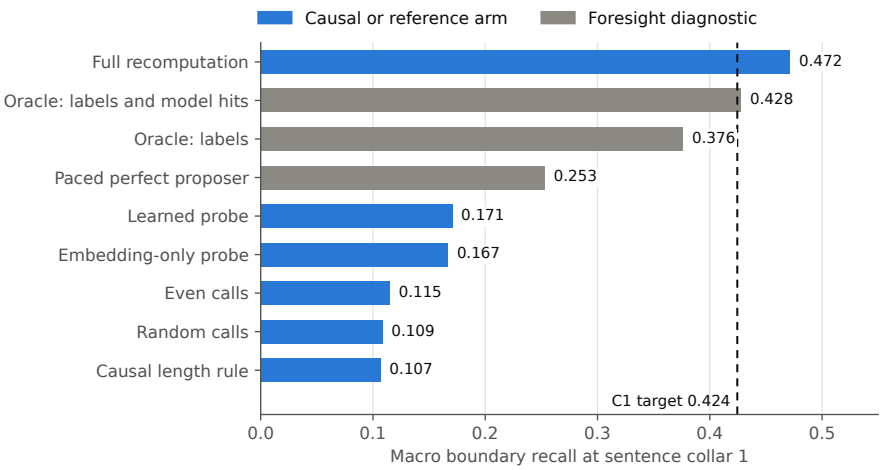


Figure 1: Macro boundary recall on YTSeg at sentence collar 1. Grey bars need knowledge that a causal policy lacks. The dashed line is the C1 target.

5. Results

5.1 Sparse recomputation falls short

Table 2: Causal sparse recomputation on YTSeg (collar 1, macro-averaged, exploratory).

Arm	Recall	Precision	F1	W=2 boundary coverage	Recall / coverage
Full recomputation	0.472	0.556	0.474	1.000	0.47
Learned probe	0.171	0.487	0.238	0.310	0.55
Embedding-only probe	0.167	0.476	0.232	0.307	0.54
Even calls	0.115	0.420	0.170	0.239	0.48
Random calls	0.109	0.411	0.163	0.225	0.48
Causal length rule	0.107	0.413	0.161	0.218	0.49

The C1 intervals across seeds spanned 0.347 to 0.377 (target 0.90), and the C2 intervals 0.051 to 0.075 (target 0.08), so both are falsified. Adding MiniSeg state to the embedding-only probe changed recall by an amount whose interval included zero (-0.003 to 0.011). Across all arms, recall is close to coverage multiplied by the full model's recall, so coverage is the dominant limit: the arms mostly differ in how many boundaries they reveal.

An earlier preregistered trigger based on a frozen embedding residual also failed: its recall was 0.139, against 0.472 for full recomputation and 0.129 for even calls. That run replayed cached model output, so we rely on the causal results above.

5.2 What limits recall (post-hoc)

Table 3: Foresight diagnostics on YTSeg at the same budget (exploratory, none deployable).

Schedule	W=2 coverage	Collar-1 recall	Ratio to full
Runner-61 oracle (maximizes collar-1 coverage)	0.227	0.138	0.29
Paced perfect proposer	0.557	0.253	0.54
Oracle using labels	0.889	0.376	0.80
Oracle using labels and model hits	0.509	0.428	0.91

C1 is only just reachable, and only with foresight of both the labels and the model's output. Under the pacing controller, even a perfect proposer reaches a ratio of 0.54, so C1 could not have been met by any signal in that design. C1's failure therefore says little about signal quality. C2 is more informative: the probe's coverage of 0.310 is well below the 0.557 that the same controller permits with a perfect proposer, so the cheap signal does leave room for improvement.

Runner-61's oracle, optimized for collar-1 coverage, reaches 0.949 of boundaries within collar 1 (interval 0.942 to 0.955), against 0.456 for even and 0.451 for random. That high figure does not translate to predictions: its W=2 coverage is only 0.227, because a boundary within the collar need not lie in the revealed window, and its recall is below the probe's. It should not be read as showing that the budget is no barrier.

By density quartile, that oracle's collar-1 coverage was 0.995, 0.981, 0.977, and 0.851. The raw q3-minus-q4 gap was 0.125, and it fell to 0.062 after adjustment (intervals about 0.045 to 0.078), below the 0.10 threshold. The adjusted gap still excludes zero. The evidence against a sharp threshold rests on the q4 term beyond a quadratic density trend (-0.012, interval including zero). The hypothesis was itself motivated by the same test data.

5.3 Placement outweighs count on YTSeg

Table 4: Collar-1 micro F1 on YTSeg by placement and count (exploratory).

Placement	Oracle K	Estimated K
MiniSeg	0.551	0.533
Cohesion	0.339	0.329
Even	0.256	0.234
Random	0.217	0.203

Knowing the true count added about 0.02, far less than the differences between placements. The count-by-placement interaction was about 0.008 with an interval including zero. MiniSeg's advantage over cohesion (0.212 at collar 1, lower bound 0.202) shrank by only 0.041 from collar 0 to collar 3 (interval 0.026 to 0.058), which failed the prespecified expectation of at least 0.15. Its advantage over even spacing shrank much more with tolerance (Table 6).

By video duration, the advantage over even spacing at collar 3 ranged from 0.011 (interval including zero) in the shortest quartile (up to 485 seconds) to 0.195 in the longest (over 1,240 seconds). This is a sensitivity analysis. Sentence collars may act as relatively wide tolerances in short videos, and the longest bin is not evidence about videos over 30 minutes.

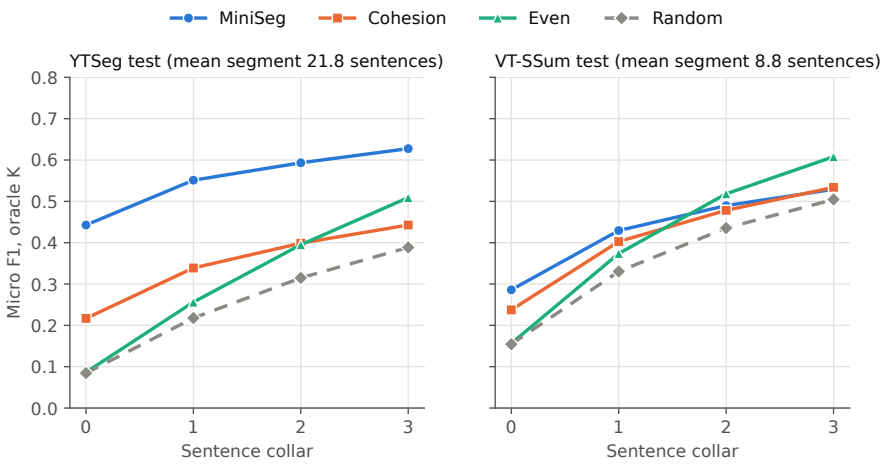


Figure 2: Micro F1 by sentence collar with oracle K. YTSeg results are exploratory and VT-SSum is the preregistered test.

5.4 Preregistered test on VT-SSum

Table 5: Preregistered claims on the VT-SSum test split (958 videos, online model).

ID	Estimate	Interval range across seeds	Threshold	Decision
P1	0.056	0.044 to 0.067	0.05	Inconclusive
P2	0.100	0.088 to 0.109	0.05	Supported
P3	0.209	0.199 to 0.218	0	Supported
P4	−0.280	−0.298 to −0.262	0	Falsified

The same rule gives the same four decisions for the secondary offline model (P1 0.059, P2 0.103, P3 0.233, P4 −0.283).

Table 6: F1 advantage by sentence collar, oracle K. Parentheses give the range of intervals across seeds.

Contrast	Collar 0	Collar 1	Collar 2	Collar 3
YTSeg: MiniSeg minus even	0.356 (0.341, 0.372)	0.295 (0.281, 0.309)	0.198 (0.184, 0.212)	0.119 (0.104, 0.132)
VT-SSum: MiniSeg minus even	0.130 (0.121, 0.138)	0.056 (0.044, 0.067)	−0.028 (−0.040, −0.017)	−0.079 (−0.089, −0.069)
VT-SSum: MiniSeg minus random	0.132 (0.122, 0.141)	0.100 (0.088, 0.109)	0.054 (0.042, 0.066)	0.025 (0.013, 0.036)
VT-SSum: MiniSeg minus cohesion	0.049 (0.041, 0.056)	0.027 (0.018, 0.035)	0.011 (0.002, 0.020)	−0.005 (−0.014, 0.004)

Table 7: Micro F1 on VT-SSum test by placement and collar (online model, oracle K, median across seeds).

Placement	Collar 0	Collar 1	Collar 2	Collar 3
MiniSeg	0.286	0.429	0.490	0.529
Cohesion	0.237	0.403	0.478	0.534
Even	0.156	0.374	0.518	0.608
Random	0.154	0.330	0.436	0.504

With its own count estimate, MiniSeg's collar-1 F1 fell to 0.094. It predicted 2,429 boundaries against 32,600 references and none in 200 videos, with a mean absolute count error of 31.5 per video and a mean Pk of 0.398. Its positive prediction rate was 0.8% against a boundary rate of 11.1% (3.2% against 4.1% on YTSeg). By sentence-count quartile (secondary, one bootstrap seed), the collar-1 advantage over even spacing was 0.020 (interval including zero), 0.042, 0.067, and 0.065, from shortest to longest.

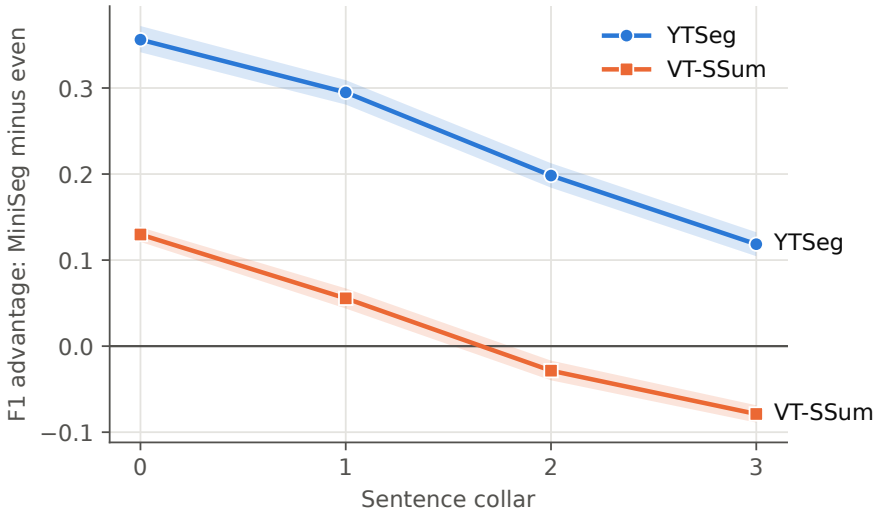


Figure 3: MiniSeg minus even spacing. Bands span the bootstrap intervals across seeds.

5.5 Peak suppression on VT-SSum (post-hoc)

Top-K selection can place boundaries next to each other, and even spacing cannot. Of consecutive MiniSeg picks, 30.7% were adjacent, against 14.7% of reference boundaries. We therefore tested label-blind peak suppression at the same K, skipping picks within a gap of an earlier pick. This choice was made after seeing test results.

Table 8: MiniSeg with peak suppression minus even spacing on VT-SSum test (exploratory, one bootstrap seed).

Selection	Collar 0	Collar 1	Collar 2	Collar 3
Top-K, as preregistered	0.130	0.056	−0.028	−0.079
Gap 1	0.121 (0.112, 0.129)	0.119 (0.109, 0.129)	0.039 (0.030, 0.048)	−0.013 (−0.021, −0.006)
Gap 3	0.117 (0.109, 0.125)	0.117 (0.107, 0.128)	0.070 (0.061, 0.079)	0.029 (0.022, 0.037)

Suppression roughly doubled the collar-1 advantage. It removed the reversal at collar 2 for both gaps and at collar 3 for gap 3. P1 to P4 stand as preregistered, but the reversal partly reflects the arm definition rather than the tolerance alone.

6. Discussion

6.1 Why the trigger failed

C1 asked for something the design almost could not deliver: only foresight of both labels and model output reaches it, and the pacing controller rules it out even with a perfect proposer. The failure of C1 therefore says more about the target and the design than about cheap signals. What the data do show is narrower. Causal cheap signals revealed about 0.31 of boundaries where the same controller allows 0.557, and their gain over random calling fell short of the prespecified 0.08. MiniSeg state added nothing we could detect beyond the embedding. A design without the pacing constraint might do differently, and we did not test that.

6.2 What transfers

The direction at tight tolerances transferred: MiniSeg beat random placement and led even spacing at collars 0 and 1. The size did not. At equal collars the advantage over even spacing fell by a factor of 2.7 at collar 0 and 5.3 at collar 1, and the advantage over cohesion fell from 0.212 to 0.027 at collar 1.

Those ratios compare unequal relative tolerances. A collar's window (2c + 1 sentences) relative to mean segment length is 0.34 for VT-SSum collar 1 and 0.32 for YTSeg collar 3. At those comparable settings the advantage is 0.056 and 0.119, a factor of about 2.1. No interval quantifies this between-dataset contrast.

We cannot separate the causes. VT-SSum differs from YTSeg in domain, in label definition (slide changes rather than chapters), and in segment length. The preregistration anticipated that wide collars favor even spacing when segments are short, but this does not by itself explain the reversal: even spacing's lead over random at collar 3 is similar on both datasets (0.121 and 0.104), and peak suppression largely removes the reversal. Slide changes only proxy topic change, so this is not simply a failure of the method.

The count result is best read as a calibration failure. One threshold, set on YTSeg, was applied under a strong base-rate shift. This does not establish that count matters more than placement in general; P4 tests this one uncalibrated rule. It does show that claims about the relative importance of count and placement can depend on the data and the calibration.

6.3 Lessons from the research process

Several errors would have changed the conclusions had they not been found:

- Three runs implemented the trigger without the W = 2 contract, so their low values reflected the instrument rather than the hypothesis.
- One probe run reported C1 of 0.830 and C2 of 0.294 by using full-stream model output at skipped positions. The causal rerun gave C1 intervals of 0.347 to 0.377.
- A margin-trigger run reported a C2 gain of 0.152. Related designs were found to be non-causal, and that run was not separately adjudicated, so we do not use it.
- One residual run, and a reconfirmation built on its output, thresholded a raw logit at 0.5 instead of comparing classes, which gave full-recomputation recall of 0.265 rather than 0.472.
- A baseline named "equal duration" was actually even sentence spacing.
- The oracle's 0.949 collar-1 coverage was initially read, including in an earlier version of this draft, as showing that the budget was no barrier. Scoring predicted recall corrected this (Section 5.2).

One inconsistency remains unexplained: two runs disagree on TextTiling-style F1 at a 10-second tolerance (0.345 and 0.245). We use neither and rely only on sentence-collar comparisons.

7. Limitations

- **Test exposure.** The YTSeg test split was opened repeatedly, and later hypotheses came from it. All YTSeg results are exploratory regardless of their manifests.
- **Post-hoc analyses.** Sections 5.2 and 5.5 were designed after results were seen, and Section 5.5 uses the VT-SSum test split. Any claim from them needs fresh data.
- **Single external dataset.** VT-SSum uses weak slide-derived labels. We have not tested against human chapters from a second source.
- **Preregistration independence.** VT-SSum hypotheses and thresholds derive from opened YTSeg results, and the experimenter saw development results before the freeze. P1 was supported on development and inconclusive on test, which shows how close it lies to the threshold.
- **Single training seed.** Bootstrap seeds do not measure variation across training runs.
- **Matching.** Greedy matching is not maximal. An unbootstrapped check suggests optimal matching would move the collar-1 contrast from 0.056 to about 0.060 and the collar-3 contrast from -0.079 to about -0.070.
- **Compute.** "Cheap" and "expensive" were not measured. The fine-tuned sentence encoder itself still runs on every sentence in these designs.
- **Review.** The cycle's gate never passed (its last score was 58, a rejection). The review of this draft was by an AI agent, and no human has reviewed it.
- **Scope.** Text only, with no audio or visual cues, although the original research request asked about them. No evidence on Korean content or videos over 30 minutes. Even spacing is by sentence index on both datasets.

8. Conclusion

Under a fixed, paced call budget, cheap causal signals did not preserve MiniSeg's boundary recall on YTSeg, and the 90% target was barely reachable even with foresight. With counts held equal, MiniSeg placement was much better than simple baselines on YTSeg. On VT-SSum the advantage over even spacing shrank, reversed at wider tolerances under top-K selection, and was overshadowed by count error from an uncalibrated threshold. Peak suppression changed the picture substantially in exploratory analysis. Chaptering results should therefore be reported across tolerances, with count separated from placement, with the boundary selection rule stated, and with the feasibility of targets checked before they are set.

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